

Towards Differentiated Governance: A Typology and Differentiated Governance Framework for County-Level Traditional Chinese Medicine Hospitals in China

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Abstract

This study addresses the heterogeneity in development potential among public county-level Traditional Chinese Medicine (TCM) hospitals in Zhejiang Province, China. Employing a sequential mixed-methods design, we conducted cluster analysis on data from 78 counties and supplemented this with 24 stakeholder interviews. Four distinct archetypes were identified: Leading Development Zones, Pressure & Challenge Zones, Steady Improvement Zones, and Foundation Building Zones. Development potential emerges from the interaction between external resource dependencies and internal adaptability. We propose a differentiated “diagnose-classify-prescribe” governance framework, advocating for tailored roles—such as “innovation catalyst” or “system stabilizer”—to move beyond uniform policies. This model offers actionable, context-sensitive strategies for integrating traditional medicine into diverse health systems locally and globally.

Research in Context

Prior studies on TCM hospitals have focused on efficiency or equity in isolation, lacking integrated frameworks linking context to capacity. This study addresses this gap by combining cluster analysis and interviews in Zhejiang to identify four distinct county archetypes. It translates this into a novel “diagnose–classify–prescribe” governance model, offering policymakers a practical tool for moving beyond one-size-fits-all approaches to TCM integration.

1. Introduction

Global health systems are confronting the dual pressures of aging populations and the rising burden of chronic non-communicable diseases [1, 2]. This has intensified the search for cost-effective, preventive, and holistic models of care [3]. Within this context, Traditional Chinese Medicine (TCM)—with its theoretical emphasis on balance, prevention, and individualized treatment—is increasingly recognized as a valuable resource [4, 5]. Its formal inclusion in the WHO’s International Classification of Diseases represents a significant step toward global legitimacy and has stimulated critical discourse on integrating traditional knowledge systems into mainstream healthcare delivery [6].

However, a persistent implementation gap often exists between high-level policy endorsement and effective local service provision. The potential of TCM to alleviate systemic pressures depends fundamentally on the robustness of its institutional carriers—particularly public TCM hospitals at the primary and secondary care levels. In China, the county represents the key administrative and operational unit for delivering healthcare to large rural and aging populations. Public county-level TCM hospitals, therefore, constitute the critical frontline in the broader challenge of integrating traditional medicine into modern health systems in a sustainable and equitable manner [7, 8].

Zhejiang Province, one of China’s most economically dynamic and demographically aging regions, presents a compelling and urgent case for investigating this issue [9]. Its pronounced internal socio-economic heterogeneity—spanning affluent metropolitan zones and less developed mountainous

counties—mirrors regional disparities observed globally. This variation creates a “natural laboratory” to examine the complex interplay between local fiscal capacity, demographic structure, and institutional development within the public hospital system—a dynamic often obscured in macro-level analyses.

To address this research gap, this study advances the field in three key aspects. First, it develops and applies an integrated analytical framework that simultaneously assesses the internal service capacity of public county-level TCM hospitals and the external socio-economic and demographic drivers of their development, moving beyond siloed evaluations. Second, it adopts a sequential mixed-methods design: establishing a quantitative typology of counties first, then systematically explaining and enriching it through qualitative insights from stakeholders, ensuring findings are both generalizable and contextually grounded. Third, and most significantly, it generates a data-driven, actionable typology of development potential for these public institutions. This typology transcends one-size-fits-all policy approaches by identifying distinct regional archetypes, each characterized by its own systemic logic and corresponding governance requirements.

The implications of this research extend beyond a single Chinese province. For policymakers and health system managers worldwide seeking to leverage traditional and complementary medicine (T&CM) within publicly-funded or publicly-managed health systems, this study offers a transferable “diagnose–classify–prescribe” model. It provides a robust methodological blueprint for assessing sub-national development potential and for formulating spatially differentiated, institutionally-aware strategies, thereby contributing to the global pursuit of more resilient, accessible, and equitable healthcare in the 21st century.

2. Literature Review

2.1. Assessing medical service capacity: from general frameworks to the public TCM hospital context

The evaluation of medical service capacity is a cornerstone of health systems research, traditionally structured around dimensions of structure, process, and outcome [10, 11]. In China, the official Hospital Grade system serves as a key structural proxy for an institution's scale and technical capability [12]. This is complemented by standard resource indicators—such as bed numbers, staffing levels, and floor area—which gauge infrastructural input [13]. Process and output metrics (e.g., outpatient/inpatient volume, bed occupancy rate) reflect operational efficiency, while outcome indicators monitor clinical quality. However, applying these general frameworks to public TCM hospitals, particularly at the county level, requires critical adaptation. The unique service model of TCM—often outpatient-centric, emphasizing preventive care (“治未病”, preventive treatment) and chronic disease management—is not fully captured by conventional acute-care-oriented metrics. This gap underscores the need for evaluation systems that respect TCM's distinctive paradigms while ensuring comparability within the broader public health system.

2.2. The socio-economic and demographic ecosystem of health resources

The development of healthcare institutions is inextricably linked to their regional context. Robust evidence confirms that local economic strength, measured by per capita GDP or fiscal capacity, is a primary determinant of public investment in health resources, directly affecting infrastructure quality and the ability to attract skilled personnel [14]. Concurrently, demographic aging—quantified by indicators like the aged dependency ratio—fundamentally reshapes healthcare demand, increasing needs for chronic disease management, rehabilitation, and long-term care [15]. The critical insight lies in the interaction of these factors: economic resources define the supply potential, while demographic structure defines the demand profile. A region with a high aging ratio but low fiscal capacity faces a fundamentally different challenge from an affluent, aging region. Yet, much of this research operates at national or provincial scales, with fewer studies drilling down to examine how these dynamics play out specifically for county-level public institutions, which are most vulnerable to local fiscal and demographic pressures.

2.3. Research on TCM development: progress and the county-level gap

Scholarly inquiry into TCM has evolved along several tracks. A substantial body of work focuses on validating the clinical efficacy of TCM for specific conditions [16]. Another stream employs operations research methods, like Data Envelopment Analysis (DEA), to evaluate the technical efficiency of TCM hospitals [17]. A third line investigates the spatial equity of TCM resource distribution, using tools like the Gini coefficient to map disparities [18]. Recently, policy-oriented research has begun exploring frameworks for the “high-quality development” of TCM [19, 20].

Despite these contributions, significant gaps persist, particularly regarding public county-level TCM hospitals. First, existing studies often operate in silos: clinical research focuses on internal validity, efficiency analyses on inputs/outputs, and equity studies on macro-distribution. Few integrate the internal capacity of these institutions (e.g., grade, beds) with the external drivers (e.g., local aging, fiscal health) of their development environment. Second, the concept of “development potential”—a forward-looking assessment of future growth trajectories based on current capabilities and contextual opportunities—remains underexplored. Third, there is a lack of typological studies that classify counties based on a holistic view of their socio-economic and institutional configurations, which is essential for moving beyond one-size-fits-all policy recommendations.

2.4. Synthesizing the gap and positioning this study

The reviewed literature establishes that medical service capacity is multi-dimensional and co-determined by internal resources and external forces. While TCM research is vibrant, it has not fully integrated these premises into a cohesive framework for assessing the development potential of public county-level TCM hospitals.

This study aims to bridge this gap. We test the following hypotheses, which explicitly link the internal world of the hospital with its external ecosystem:

H1: A county's fiscal self-sufficiency rate positively correlates with the grade level of its public TCM hospital.

H2: Counties with higher aging levels will demonstrate a stronger orientation of their public TCM hospitals towards specialized geriatric and chronic disease management services.

H3: Counties with higher per capita consumption expenditure will have a higher density of TCM service resources (e.g., TCM physicians per 1,000 population) within their public hospital system.

H4: The development capacity of public county-level TCM hospitals exhibits a spatially clustered pattern that correlates with regional socio-economic development levels.

2.5. Theoretical framework: an integrated resource dependence and complex systems lens

The evaluation index system (Table 1) was designed to operationalize these two core dimensions: “Existing Service Capacity” proxies foundational adaptive capacity, while “Regional Development Support Potential” reflects the structure of the Syndicated Resource Base.

The initial hypotheses (H1-H4) were posited to investigate fundamental bivariate relationships. Building upon this groundwork, we developed the more integrative and dynamic Resource Dependence–Complex Systems (RD-CS) framework. From this framework, three core theoretical propositions (P1–P3) were derived to guide the subsequent mixed-methods empirical analysis.

To theorize the interplay between external resources and institutional development, we integrate Resource Dependence Theory (RDT) and a Complex Adaptive Systems (CAS) perspective into a unified RD-CS Framework (Fig. 1). This synthesis addresses a core tension: while RDT elucidates how organizations navigate external dependencies and power dynamics [21, 22], its premise of managerial strategic choice is profoundly constrained within China’s hierarchical public hospital system. A CAS perspective [23] complements this by foregrounding system-level emergence—the phenomenon whereby macro-level patterns (e.g., stable resource configurations) arise from the nonlinear interactions and bounded strategic adaptations of multiple agents [24]. This integrated lens moves beyond attributing outcomes solely to either resource determinism or managerial agency.

Our integrated RD-CS Framework posits that the development potential of a public county-level TCM hospital is an emergent system-level outcome. It arises not from external resources or internal agency alone, but from the dynamic and iterative interaction between two core, interconnected forces:

1. The Structure of External Resource Dependencies (The Syndicated Resource Base): This constitutes the opportunities and constraints for the hospital—a tightly coupled constellation of local fiscal capacity,

demographic demand, and political-institutional support.

2. The Adaptive Capacity of the Hospital as an Agent within a Complex System: This refers to the institution's bounded capability to perceive its environment, interpret dependencies, and mobilize resources. Crucially, this capacity is shaped and constrained by system properties like historical legacies (path dependency), policy rigidity, and competitive-cooperative relationships with other agents (e.g., larger metropolitan hospitals).

The core of our framework lies in this interaction: the Syndicated Resource Base sets the initial conditions and boundaries for action, while the hospital's Adaptive Capacity determines its strategic response within those bounds. Their interplay, mediated by reinforcing or balancing feedback loops (e.g., talent flows, reputation cycles), leads the system to converge toward distinct, stable states ("attractors"). This perspective explains why uniform policies often fail and why context-specific governance is essential.

From this framework, we derive three core, testable propositions that guide our empirical investigation:

P1 (Syndicated Resource Base Proposition): The development trajectory of a county-level TCM hospital is more strongly influenced by the coherence and strength of the local syndicated resource base (the coupling of economic, demographic, and institutional resources) than by any single resource factor in isolation.

P2 (Adaptive Capacity Asymmetry Proposition): Hospitals facing similar resource dependencies will exhibit divergent development potentials based on differences in their internal adaptive capacity, which is historically constituted and modulated by system-level feedback (e.g., reputation loops, talent flow patterns).

P3 (System Attractor Proposition): The county-level TCM service system will tend towards a limited set of stable configurations or "system attractors" (e.g., high-resource/high-adaptation; high-demand/low-resource trapped). These archetypes represent stable equilibria resulting from specific alignments or misalignments between the syndicated resource base and institutional adaptive capacity.

This framework moves beyond siloed explanations. It allows us to interpret why formally similar hospitals in different counties follow divergent paths, and why simplistic, uniform resource injections often fail. It provides the theoretical foundation for our mixed-methods design: the quantitative analysis identifies the emergent attractor states (clusters), while the qualitative inquiry probes the mechanisms of structured agency—the specific dependencies, constraints, and adaptive maneuvers—that produce and sustain these states.

3. Methodology

This study employed a sequential explanatory mixed-methods design [25, 26] to holistically assess the development potential of public county-level TCM services in Zhejiang Province. Guided by the integrated RD-CS framework, the research design strategically combined quantitative and qualitative approaches.

This allowed for the establishment of a generalizable typology through statistical analysis, which was subsequently explained, contextualized, and enriched through in-depth stakeholder insights. The sequential design is illustrated in Fig. 2.

3.1. Data Sources and Sample

3.1.1. Quantitative data and sample definition

A comprehensive panel dataset was initially constructed for all 90 county-level administrative units in Zhejiang Province for the year 2024, covering socio-economic and demographic variables.

Institutional inclusion criteria and analytical sample: The core objective of quantitatively assessing development potential requires the pre-existence of the service entity to be assessed. Therefore, we applied a key inclusion criterion: the county-level unit must possess at least one established public TCM or integrated Chinese-Western medicine hospital. Based on this criterion, 78 of the 90 units formed our primary analytical sample. The remaining 12 units without such an institution were excluded from the subsequent cluster analysis, as the concept of “development potential” for a specific hospital is not applicable where none exists. This approach ensures our typology is grounded in the operational reality of service provision.

TCM institutional data: Variables including hospital grade, number of approved beds, and floor area for all public county-level TCM hospitals and integrated Chinese-Western medicine hospitals were sourced from official provincial health administration registries (as of November 2025). These data represent the institutional outcomes shaped by the preceding regional context.

Socio-economic and demographic data for 2024—including resident population, fiscal self-sufficiency rate (calculated as $[\text{General Public Budget Revenue} / \text{General Public Budget Expenditure}] \times 100\%$), the municipal-level proportion of the population aged 60 and above (a proxy for county-level aging pressure, justified below), and per capita consumption expenditure (specifically, the per capita disposable consumption expenditure of urban residents from local Statistical Yearbooks)—were obtained from official sources. For the one instance of missing consumption expenditure data, the 2024 city-level average for Huzhou was imputed.

This cross-sectional pairing is designed to identify the systemic “attractor” states of county-level TCM ecosystems, where the 2024 regional context shapes the institutional outcomes observed in 2025. This approach is suited to our aim of diagnosing current typologies rather than modeling temporal causality.

Measurement Note: A recognized consideration in our measurement strategy concerns the indicator for aging pressure. Publicly available, fine-grained aging data at the county level are not systematically published in Zhejiang. Therefore, we employed the municipal-level elderly population proportion as a proxy. We explicitly acknowledge this as a limitation that may smooth over intra-city variations. Nonetheless, the municipal-level indicator is a defensible proxy for the broader regional aging pressure

faced by counties within that administrative and economic sphere. To mitigate this limitation and validate our interpretation, findings reliant on this proxy (e.g., the high demand characterizing the “Pressure & Challenge Zone”) were rigorously triangulated with qualitative interview data, where stakeholders explicitly detailed local service demand profiles and patient demographics..

All data were matched using standardized administrative codes, cross-verified, and subjected to logical consistency checks.

3.1.2. Data integration for units with multiple hospitals

Within the analytical sample of 78 units, two districts (Gongshu and Linping) each possessed two independent public TCM hospitals. To maintain each county-level unit as a single, unique observation in statistical analyses, a “primary hospital selection” rule was applied. For such units, the hospital with the highest official grade was selected as the representative institution. If grades were equal, the hospital with the larger number of approved beds was chosen. This approach is theoretically justified as the highest-grade hospital typically represents the peak service capability, commands core resources, and functions as the regional benchmark, aligning with the study’s focus on assessing “development potential”.

3.1.3. Qualitative data: semi-structured interviews

To triangulate and deepen the quantitative findings, semi-structured interviews were conducted. Using purposive maximum-variation sampling, 24 key informants were selected from counties identified as representative of each cluster in the quantitative typology. The sample included:

Senior managers and department heads of public TCM hospitals (n = 8)

Frontline TCM physicians (n = 8)

Officials from local Health Commissions (n = 4)

Community representatives and long-term patients (n = 4)

Interviews, lasting 45–60 minutes, were audio-recorded, transcribed verbatim, and managed using NVivo 14 software, with all recordings kept strictly confidential and not for public release to ensure participant privacy. The study was approved by the Ethics Review Committee of Zhejiang Pharmaceutical University. All participants provided written informed consent.

3.2. Evaluation index system

The evaluation index system was constructed and refined based on a previously validated framework derived from a Delphi expert consultation process for assessing county-level TCM hospital service capacity [27]. For province-wide applicability, the framework was adapted to prioritize indicators with publicly available, standardized data across all counties.

The final parsimonious system comprises six indicators across two primary dimensions (Table 1), maintaining alignment with expert-validated constructs while ensuring empirical feasibility. The “Existing Service Capacity” dimension captures the internal resource base of the hospital, while the “Regional Development Support Potential” dimension captures key external dependencies identified by the RD-CS framework.

Table 1
Evaluation index system for TCM service development potential.

Primary Dimension	Secondary Dimension	Specific Indicator	Measurement & Rationale
Existing Service Capacity	Institutional Level	Hospital Grade ¹	Quantified (3A = 5, 3B = 4, 2A = 3, 2B = 2, 1 = 1). A composite measure of comprehensive capability.
	Resource Scale	Approved Beds	Absolute number. Direct measure of inpatient service capacity.
		Floor Area (m ²)	Absolute number. Standard measure of physical infrastructure scale.
Regional Development Support Potential	Economic Foundation	Per Capita Consumption Expenditure (RMB)	Continuous. Reflects regional economic strength and individual purchasing power for healthcare.
		Fiscal Self-Sufficiency Rate (%)	Ratio of local fiscal revenue to expenditure. Measures governmental financial capacity for public health investment.
	Market Demand & Base	Municipal-Level Elderly Population Proportion (%)	% of population aged 60+. Core demographic driver for TCM demand (chronic/palliative care).
		Resident Population (10,000 persons)	Absolute number. Fundamental indicator of potential service market size.

¹Ungraded hospitals were assigned a value of 1 based on their basic scale, a treatment validated through sensitivity analysis.

3.3. Data analysis methods

3.3.1. Quantitative analysis

Data analysis and visualization were performed using OriginPro (Version 2024) for statistical computing and ArcGIS (Version 10.8) for geospatial mapping.

Descriptive and correlation analysis: Descriptive statistics and bivariate Pearson correlation analyses were conducted in OriginPro to provide an overview of the variables and to offer preliminary tests of the hypothesized relationships.

Cluster analysis (K-means): The K-means cluster analysis, serving as the core method for typology development, was performed using OriginPro. The six standardized indicators (Z-scores) served as inputs. The optimal number of clusters ($k = 4$) was determined by simultaneously evaluating the elbow method (plot of within-cluster sum of squares) and the average silhouette width, which indicated a solution with good internal cohesion and external separation.

To validate the robustness of this four-cluster solution, we conducted sensitivity analyses with alternative cluster numbers. A three-cluster solution resulted in a lower average silhouette width and failed to meaningfully differentiate the dynamics observed in what became the ‘Steady Improvement’ and “Leading Development” zones. A five-cluster solution produced an additional cluster with very small membership and indistinct, theoretically uninterpretable characteristics. Thus, the four-cluster model was retained as the most parsimonious and substantively meaningful typology. Based on this validated configuration, the K-means algorithm was then applied to partition the 78 counties into the four distinct clusters.

Geospatial visualization: The geographic distribution of the resulting four clusters was visualized and mapped using ArcGIS 10.8 to create the final spatial presentation (Fig. 3).

3.3.2. Qualitative analysis

Interview transcripts were analyzed using a reflexive thematic analysis approach. After familiarization, two researchers independently conducted initial coding, iteratively developed themes through discussion, and refined them in relation to the quantitative clusters. The analysis aimed to explain the mechanisms behind the clusters (e.g., how resource scarcity manifests) and to uncover stakeholders’ strategic adaptations.

3.3.3. Integrated analysis

The final stage involved a comparative case analysis. Representative counties from each quantitative cluster were selected as cases. Their quantitative profiles were juxtaposed with the rich qualitative themes extracted from their corresponding interviews. This integration provided a nuanced, contextually grounded understanding of the development dynamics for each archetype, fulfilling the core objective of the mixed-methods design to explain and elaborate the quantitative typology.

4. Results

4.1. Descriptive statistics

Descriptive statistics for the core variables across the 78 county-level units with established public TCM hospitals are presented in Table 2. The data reveals substantial heterogeneity in both institutional capacity and regional contexts, confirming the necessity of a typological approach. The average hospital grade approached the “2A” level (quantified mean = 3.15), but with a wide range (1 to 5). Similarly, approved beds and floor area showed significant variation, indicating disparities in infrastructure scale.

The socio-economic indicators also displayed considerable spread, particularly fiscal self-sufficiency (14.91% to 153.14%) and per capita consumption expenditure, underpinning the diverse developmental ecosystems these hospitals operate within.

Table 2
Descriptive statistics of core variables for counties with TCM hospitals (N = 78).

Variable	Mean	Standard Deviation	Minimum	Maximum
Hospital Grade (Quantified)	3.15	1.17	1	5
Approved Beds	420.05	257.79	43	1200
Floor Area (10,000 m ²)	5.07	3.67	0.30	15.00
Per Capita Consumption Expenditure (RMB)	43824.60	8777.93	23984	59895
Fiscal Self-Sufficiency Rate (%)	63.24	31.15	14.91	153.14
Municipal-Level Elderly Population Proportion (%)	27.50	3.15	23.15	36.15
Resident Population (10,000)	73.72	44.09	6.40	216.40

4.2. Correlation analysis

It should be noted that the indicator “Municipal-Level Elderly Population Proportion” is measured at the municipal level, while all other variables are county-level. This discrepancy necessitates cautious interpretation of its correlations.

In summary, while H1 was not supported by the data, the analysis revealed the critical coupling within the regional socio-economic ecosystem. The bivariate correlation analysis yielded a nuanced picture of the relationships among core variables, with several findings deviating from the initial hypotheses while revealing strong systemic linkages.

Contrary to the postulated hypothesis (H1), Hospital Grade did not show a significant correlation with direct economic metrics such as Per Capita Consumption Expenditure ($r = 0.021$, $p = 0.854$) or Fiscal Self-Sufficiency Rate ($r = 0.182$, $p = 0.110$). Instead, it demonstrated a significant moderate positive correlation with Resident Population ($r = 0.412$, $p < 0.001$), suggesting that hospital tier is more closely associated with regional service scale than with short-term economic indicators.

The most salient finding is the exceptionally strong internal coupling within the regional socio-economic ecosystem. A very strong positive correlation was observed between Per Capita Consumption Expenditure and Fiscal Self-Sufficiency Rate ($r = 0.787$, $p < 0.0001$). Furthermore, both of these economic capacity indicators showed strong positive correlations with Resident Population (Consumption-Population: $r = 0.629$, $p < 0.0001$; Fiscal-Population: $r = 0.741$, $p < 0.0001$). This triadic strong coupling

indicates that more populous counties tend to possess simultaneously higher economic vitality and greater fiscal autonomy, forming a consolidated resource base.

Regarding infrastructure, Approved Beds showed a significant but moderate positive correlation with Resident Population ($r = 0.389$, $p < 0.001$), providing partial but not decisive support for the influence of market scale. As expected, Approved Beds and Floor Area were highly correlated ($r = 0.863$, $p < 0.0001$), confirming internal consistency in physical resource allocation. Hospital Grade was also strongly linked to these scale variables (with Beds: $r = 0.792$, $p < 0.0001$; with Area: $r = 0.589$, $p < 0.0001$).

Finally, a weak positive correlation was observed between the municipal-level aging rate and county-level fiscal self-sufficiency ($r = 0.308$, $p = 0.006$). No significant association was found with county population size ($r = 0.119$, $p = 0.301$).

In summary, while H1 was not supported, H4 was clearly validated by the spatially clustered patterns revealed in the subsequent analysis (Section 4.3). H2 and H3 could not be directly tested with the available indicator set but informed the broader interpretation of the typology.

4.3. Cluster analysis: A four-type typology

Guided by the RD-CS framework, which posits that county-level TCM hospital systems tend toward a limited set of stable equilibria (“systemic attractors”) based on specific resource-ability configurations, our K-means cluster analysis provided empirical validation for this proposition. The 78 county-level units were clearly partitioned into four internally homogeneous yet mutually distinct clusters. Each cluster represents a unique systemic attractor, revealing a specific pattern of coupling or mismatch between the “Syndicated Resource Base” and institutional adaptive capacity (see Fig. 3 and Table 3).

Table 3
Characteristics and underlying dynamics of the four systemic attractors.

Systemic Attractor (Cluster)	Representative Counties	Quantitative Profile: Syndicated Resource Base & Institutional Capacity (Mean Values)	Mechanistic Elucidation: Adaptive Strategies & Systemic Feedback (From Qualitative Evidence)
I: Leading Development Zone	Xiaoshan, Yiwu, Zhuji, Wenling	• Strong Syndicated Resource Base: Highest Per Capita Consumption Expenditure (¥57.8K) and Fiscal Self-Sufficiency Rate (115%)	Theme: Strategic Innovation & Regional Hub Aspiration. Hospitals exhibit high strategic agency, actively constructing a positive feedback loop of technical reputation. One director stated: “Our strategy is to develop flagship TCM departments (e.g., rehabilitation, preventive care) to attract patients region-wide, aiming to establish a specialty hub.” (Hospital Manager, Interview #1). Their development logic has shifted from scale expansion to quality-driven branding.
II: Pressure & Challenge Zone	Fuyang, Tonglu, Ninghai, Ruian, Dongyang	• Fragile Syndicated Resource Base: Weak economic (Consumption: ¥42K) and fiscal (Self-Sufficiency: 45%) foundations • Constrained Institutional Capacity: Relatively low Hospital Grade (2.5) and Beds (180) • High External Demand Pressure: Highest Elderly Proportion (30.1%)	Theme: System Overload & the Negative Cycle of Resource Siphoning. Characterized by a sharp “high-demand-low-resource” contradiction, their adaptive capacity is systemically constrained. A practitioner explained: “We are inundated with elderly chronic patients but lack capacity. The bigger challenge is proximity to [metropolitan centers]—patients and young doctors are easily drawn away.” (TCM Practitioner, Interview #14). This fosters a lock-in effect of “demand backlog → resource (talent, patient) outflow → further eroded capacity”.
III: Steady Improvement Zone	Shangcheng, Haishu, Wuxing, Luqiao	• Highly Developed Syndicated Resource Base: Exceptionally high Per Capita Consumption (¥59.9K) and strong Fiscal Self-Sufficiency (85%) • Moderate Foundational Institutional Capacity: Medium Hospital Grade (3.2) and Beds (290)	Theme: Niche Competition & Service Integration. Located in urban cores, their agency manifests in differentiation and embedding strategies. A department head shared: “We cannot compete on scale with provincial hospitals. Our focus is on deeply integrating TCM into chronic disease management pathways within the city’s healthcare ecosystem.” (Department Head, Interview #7). They build adaptive capacity by enhancing their “indispensability”

Systemic Attractor (Cluster)	Representative Counties	Quantitative Profile: Syndicated Resource Base & Institutional Capacity (Mean Values)	Mechanistic Elucidation: Adaptive Strategies & Systemic Feedback (From Qualitative Evidence)
		• Lowest Aging Rate (24.2%), yet faces intense competition	within the modern healthcare system.
IV: Foundation Building Zone	Kaihua, Suichang, Qingtian, Daishan	• Comprehensively Weak Syndicated Resource Base: Lowest economic (Consumption: ¥38K) and fiscal (Self-Sufficiency: 38%) indicators • Weakest Foundational Institutional Capacity: Lowest Hospital Grade (1.8) and Beds (95) • High Aging Rate (27.5%)	Theme: Basic Survival Crisis & Capacity Hollowing-Out. The system resides in a primary attractor state, where building adaptive capacity faces fundamental constraints. A manager stated: “The core vicious cycle is: low grade and remote location hinder talent recruitment and retention, which directly limits service quality and development.” (Hospital Manager, Interview #4). The central challenge is breaking the poverty trap of “resource scarcity → net talent outflow → capacity stagnation”.

4.4. Unpacking the black box: qualitative insights into adaptive capacity and systemic feedback

The qualitative interview analysis uncovered the micro-level mechanisms sustaining the aforementioned systemic attractors, particularly the “structured agency” practiced by hospitals as active agents and the key feedback loops driving system dynamics. The findings resonate deeply with the three theoretical propositions.

First, the management strategies toward “resource dependence” confirm the fundamental conditioning role of the Syndicated Resource Base (aligning with P1). Hospital administrators’ strategic awareness and scope of choice are systematically shaped by the strength of their local resource base. In Leading Zones, discussions center on “how to compete for provincial key specialty projects” (proactive planning); in Pressure Zones, they shift to “how to secure guaranteed personnel funding from county finance” (struggle for survival). This empirically demonstrates that the structure of the Syndicated Resource Base sets the initial boundaries and tenor of agency.

Second, the construction of hospitals’ “adaptive capacity” reveals divergent pathways (aligning with P2). Despite constraints, hospitals within different attractors develop distinct core adaptive logics. Leading Zone hospitals shape positive feedback loops by building “technical high grounds”; Steady Improvement Zone hospitals consolidate niche markets through “service embedding” within existing ecosystems; meanwhile, a significant portion of agency in Pressure and Foundation Zones is consumed by “defensive” strategies coping with resource drainage and maintaining basic operations. This indicates

that divergences in adaptive capacity serve as a critical mediating variable leading to developmental bifurcation, even under similar resource dependencies.

Finally, the interviews clearly identify core feedback loops that maintain or alter system states (aligning with P3). The two most prominent loops are:

The Negative Feedback Loop of Talent Mobility: Pervasive in Pressure and Foundation Zones, articulated as a vicious cycle: “poor remuneration → loss of key staff → department deterioration → declining revenue → further worsened remuneration.” A health official acknowledged: “The first choice for our newly licensed TCM physicians is to move to Hangzhou or Ningbo.” (Health Official, Interview #18).

The Positive Feedback Loop of Brand and Trust: Clearly observed in Leading Zones, manifested as “characteristic efficacy → patient word-of-mouth/referral → increased service volume → growth in revenue and research resources → further attraction of talent and enhancement of techniques.” A patient representative noted: “People here trust the TCM in this hospital; they are willing to come even from farther away.” (Patient Representative, Interview #22).

In summary, the qualitative findings transform static quantitative features into dynamic systemic processes. For instance, the high aging rate (a quantitative indicator) in the “Pressure & Challenge Zone” materializes in interviews as the “daily overload of geriatric clinics” (qualitative reality), while weak fiscal indicators correlate with management dilemmas like “the inability to update outdated TCM equipment.” These micro-level perceptions, decisions, and interactions constitute the very foundation upon which the macro-level “systemic attractors” form and persist, strongly corroborating the nature of “development potential” as an emergent property.

5. Discussion

This study, by integrating a theoretically-grounded typology with mechanism-based qualitative insights, transcends a simple descriptive classification. It reveals that the divergent development pathways of public county-level TCM hospitals are not random but represent distinct, stable systemic configurations. The following discussion interprets these configurations—conceptualized as systemic attractors—through the lens of our RD-CS framework. We explicitly examine how the empirical findings speak to our three theoretical propositions, derive nuanced governance implications, and candidly address the study's limitations.

5.1 Interpreting attractors through the RD-CS Lens: propositional analysis

5.1.1 Proposition 1 (syndicated resource base): the primacy of coupled resources over isolated factors

The strong empirical coupling among economic vitality, fiscal capacity, and population scale within the quantitative clusters provides robust support for P1. The development trajectory of a hospital is decisively shaped not by any single resource but by the coherence and strength of this local “Syndicated Resource Base.” For instance, Leading Zones exhibit a consolidated, high-strength base, enabling strategic freedom. Conversely, Pressure Zones suffer from a fragile base (low economy/fiscal) mismatched with high demographic demand, creating systemic strain. This finding challenges policy approaches that target isolated inputs (e.g., merely adding beds) and underscores the need for holistic, context-sensitive assessments of a region's total resource ecosystem.

5.1.2 Proposition 2 (adaptive capacity asymmetry): agency within bounded structures

P2 is vividly illustrated by the divergent strategic postures observed within similar quantitative constraints. The decoupling of hospital grade from immediate economic metrics (as seen in the correlation analysis) and the qualitative evidence of varied strategic choices confirm that adaptive capacity is a critical mediator. Steady Improvement Zones, despite not leading in scale, leverage integration and niche specialization—a form of structured agency—to thrive in competitive urban ecosystems. In contrast, Foundation Building Zones exemplify severely bounded agency, where adaptive energy is consumed by basic survival. This affirms our core theoretical tenet: development potential emerges from the interaction between the external resource structure and the internal, historically shaped capacity to adapt.

5.1.3 Proposition 3 (system attractor): emergent stability and path dependency

The four-cluster solution offers compelling empirical evidence for P3. Each attractor represents a quasi-stable system state maintained by dominant feedback loops, explaining the persistence of development patterns.

Leading Zones are locked in a virtuous cycle (a reinforcing loop): Strong resources → investment in quality/specialization → enhanced reputation and revenue → further resource accumulation and attraction of talent.

Pressure Zones are trapped in a vicious cycle (a balancing loop that reinforces stagnation): High demand & resource scarcity → overwhelmed services and outmigration of talent/patients → diminished capacity and revenue → increased difficulty in improving resources and retaining staff.

The qualitative identification of these specific loops (e.g., “siphon effect”, talent drain, brand building) provides the micro-mechanisms for the macro-level stability of these attractors, demonstrating path dependency and the non-linear nature of system change.

5.2 Theoretical implications: advancing the RD-CS synthesis

This study moves the RD-CS framework from a conceptual lens to an empirically informed analytical tool. First, it operationalizes the “Syndicated Resource Base” as a measurable construct crucial for understanding public sector dependencies. Second, it clarifies that “adaptive capacity” in complex public systems is not merely managerial skill but a systemic property influenced by history, location, and institutional legacies. Finally, by mapping empirical clusters to theoretical attractors and their sustaining loops, we provide a template for a “middle-range theory” that connects abstract complexity concepts to concrete, observable policy domains. This bridges the often-separated worlds of organizational theory and health systems research.

5.3 Governance implications: from uniform inputs to differentiated system intervention

The attractor-based diagnosis necessitates a paradigm shift from homogeneous resource allocation to targeted system modulation. Governance must aim to alter the critical feedback loops defining each attractor.

For Leading Zones (Virtuous Cycle): Governance should adopt a stance of “Strategic Empowerment and Platform Building”. This entails granting greater autonomy in service innovation and pricing to fuel the reinforcing loop, while mandating regional mentoring roles to diffuse benefits.

For Pressure & Challenge Zones (Vicious Cycle): The required governance role is that of a “Systemic Stabilizer and Circuit-Breaker”. Policy must directly address the core negative feedback loops.

As a “Resource Injector & Stabilizer”: Provide asymmetric, earmarked fiscal transfers from provincial or central levels to ensure baseline operational security and fund targeted capacity-building programs, breaking the “scarcity → outflow” trap.

As a “Systemic Connector”: Formally integrate these hospitals into regional medical alliances or tiered referral systems dominated by leading urban hospitals. Mandate structured talent rotation, remote consultation, and standardized referral pathways from core urban hubs. This transforms the uncontrolled “siphon effect” into a managed “supportive circulatory system”, allowing resource flow back into the county.

Leverage Existing Policy Tools: This approach can be operationalized through deepening tight-knit county medical communities reforms, strengthening pairing assistance programs with binding deliverables, and designing DRG/DIP that incentivize retaining and managing chronic patients locally rather than outright referral.

For Steady Improvement Zones (Niche Competition): Policy should serve as an “Integrator and Legitim�er”. Facilitate and incentivize the formal integration of TCM protocols into standard chronic care pathways within medical consortia, solidifying their valued niche.

For Foundation Building Zones (Poverty Trap): Policy must provide “Foundational Support and Trusteeship.” Ensure a basic operational safety net through provincial finance. Explore direct trusteeship by leading hospitals to implant management and technical capacity, initiating a new path.

Implementing the Differentiated Approach: Operationalizing this framework requires a two-step process. First, health authorities can apply the diagnostic index system (Table 1) and clustering methodology to map their own sub-national units into analogous archetypes. Second, governance resources and policy tools must then be deliberately aligned with the specific roles (e.g., “Systemic Connector”, “Integrator”) prescribed for each archetype, moving from uniform input allocation to targeted system modulation.

5.4 Limitations and future research

While our mixed-methods approach and theoretical framework strengthen the analysis, key limitations must be acknowledged.

Causality and Dynamics: The cross-sectional design identifies attractors but cannot model the transition pathways between them. Future longitudinal or simulation studies (e.g., System Dynamics) are needed to test the long-term impact of the proposed differentiated interventions.

Measurement Constraints: The use of municipal-level aging data as a proxy and the lack of granular hospital operational data (e.g., TCM-specific service mix, patient flow origins) limits precision. Future research should seek to incorporate finer-grained indicators to refine the typology.

Contextual Specificity: The attractors are identified within Zhejiang’s specific socio-economic and policy context. The manifestation of the “siphon effect” or the precise resource coupling may differ elsewhere. The primary contribution of this study is the diagnostic “diagnose-classify-prescribe” logic and the RD-CS analytical framework, not the specific four-type map itself. Applying this framework to other regions in China and globally is the essential next step for validating and refining its general utility.

Agency-Capacity Nexus: While we identify adaptive capacity as crucial, further research is needed to dissect its components (e.g., leadership, organizational learning) and how they can be cultivated under different resource constraints.

6. Conclusion

This study demonstrates that the development potential of county-level TCM hospitals is an emergent property of complex, path-dependent systems. By integrating RD-CS theories, we have advanced beyond descriptive categorization to provide a mechanism-based explanation for regional disparities. The identification of four systemic attractors, each with its own logic of resource coupling, adaptive agency, and sustaining feedback loops, offers a powerful diagnostic tool. For policymakers, the imperative is to abandon one-size-fits-all approaches and embrace differentiated, system-aware governance strategies that are precisely calibrated to disrupt vicious cycles and reinforce virtuous ones. The RD-CS framework and the attendant methodology presented here offer a replicable model for achieving context-sensitive

integration of traditional medicine into modern health systems, both in China and in other settings grappling with similar challenges of equity and systemic development.

Declarations

Conflict of interest

The authors declare no conflicts of interest.

Ethics declaration

The study was approved by the Ethics Review Committee of Zhejiang Pharmaceutical University. All participants provided written informed consent.

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Author Contribution

Conceptualization, formal analysis, funding acquisition, methodology, preparation, Hanhua Wang; data curation, investigation, resources, writing-original draft, Yang Yang; supervision, validation, visualization, writing-review and editing, Yingying Ji; project administration, software, Jingjing Huang.

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Data Availability

The data underlying this article are available in the article and in its online supplementary material; however, the interview transcripts are not publicly shared to protect the privacy of the participants.

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Figures

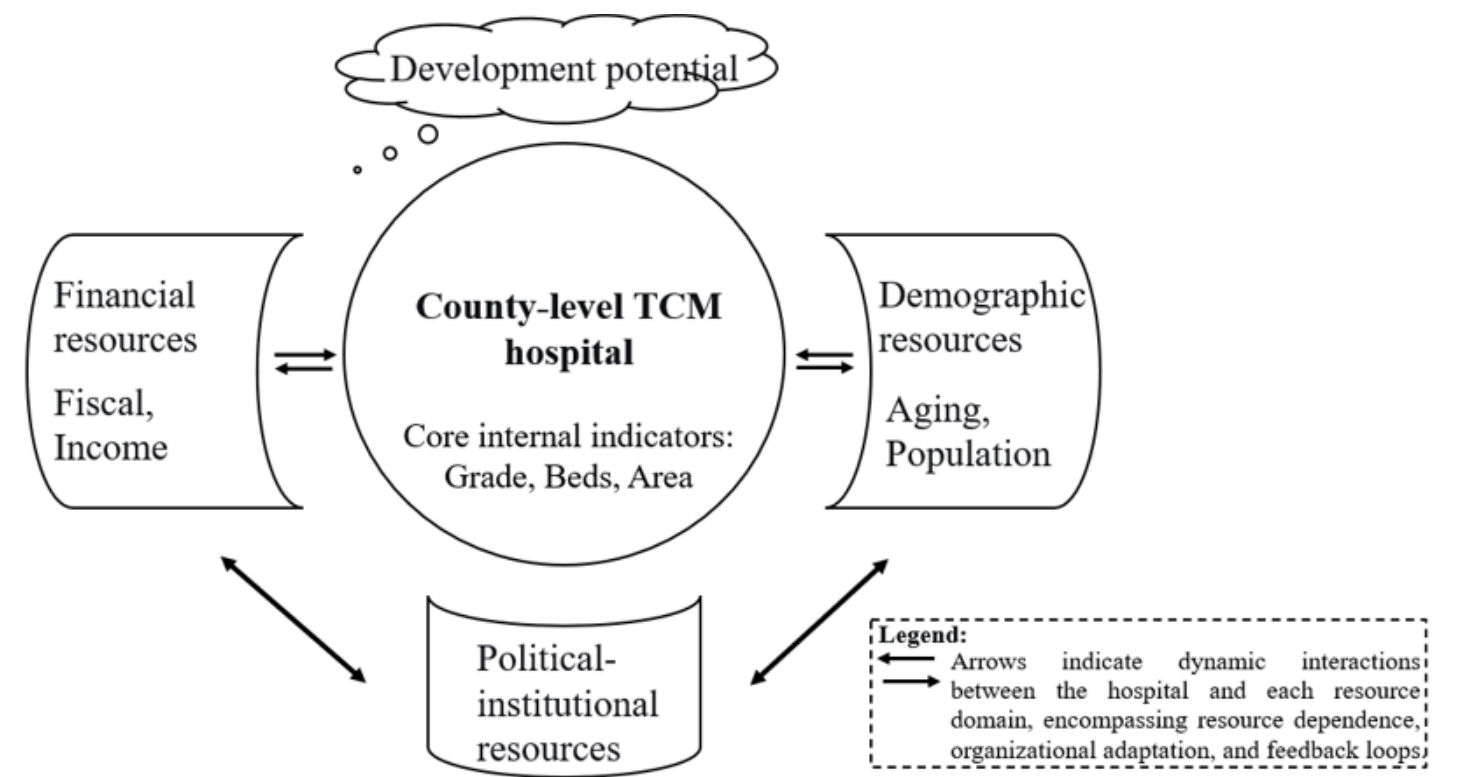


Figure 1

The Integrated RD-CS analytical framework.

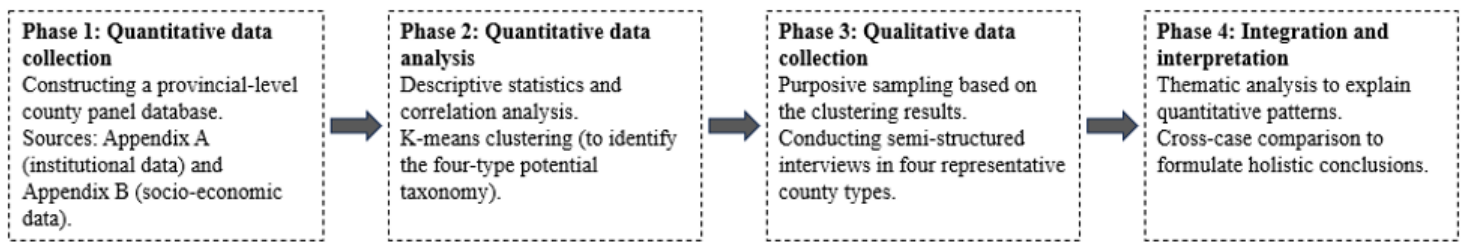
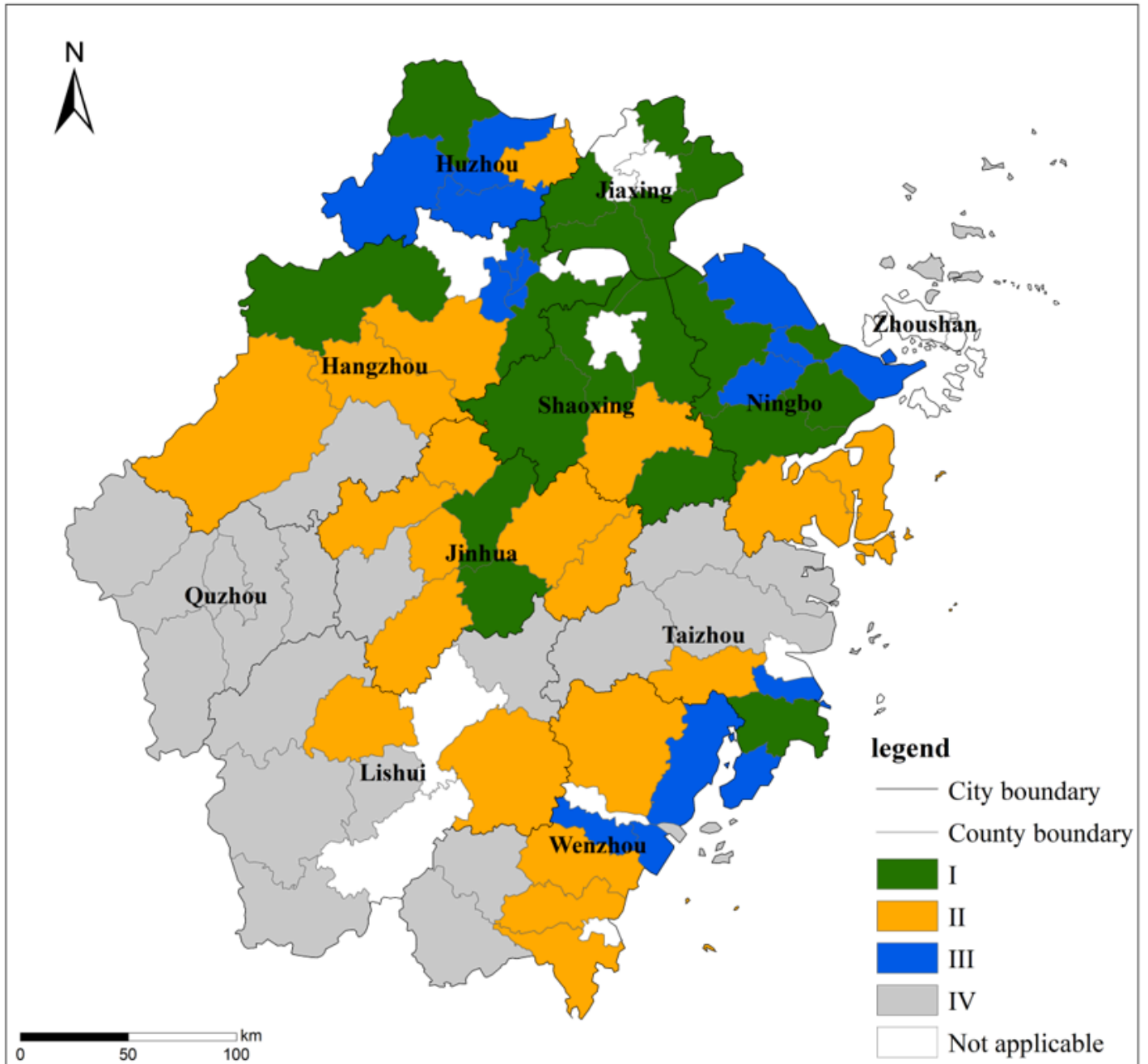


Figure 2

Sequential mixed-methods research design.



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Figure 3

Geographic distribution of the four systemic attractors in Zhejiang's county-level TCM hospital ecosystem.

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